

POULTRY FARMING OPERATIONAL GUIDELINE

Standard Operating Procedures for Broiler & Layer Production Management

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Target Audience: Farm Managers & Technicians

Scope: Commercial Poultry Operations

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Approved By: Agricultural Advisory Board

1. Introduction & Biosecurity Protocols

Biosecurity is the foundation of successful poultry farm management. A strict biosecurity program minimizes the entry and spread of infectious disease agents, ensuring optimal flock performance and profitability.

CRITICAL BIOSECURITY RULES

- Access Control:** Restrict all unauthorized personnel and vehicles from entering the farm perimeter.
- Sanitation:** All visitors and farm staff must utilize footbaths with fresh disinfectant at every house entrance.
- All-In / All-Out:** Operate strict batch production systems to allow full house cleaning and disinfection between cycles.
- Pest Control:** Maintain active rodent and wild bird exclusion programs around storage and housing areas.

2. Environmental Management & Housing

Proper environmental control ensures optimal growth rates, feed conversion efficiency, and flock welfare. Ventilation, temperature, and lighting must be carefully monitored daily.

Temperature & Ventilation Guidelines

Chicks require a warm, stable environment during brooding. Gradually reduce house temperature as birds develop feather coverage, as detailed in Table 1.

Age (Days / Weeks)	Target Temp (°C)	Target Temp (°F)	Relative Humidity (%)	Ventilation Type
Day 1 – Day 3	32 – 34 °C	90 – 93 °F	60 – 70 %	Minimum (Air Exchange Only)
Day 4 – Day 7	30 – 32 °C	86 – 90 °F	60 – 70 %	Minimum Ventilation
Week 2 (Days 8–14)	28 – 30 °C	82 – 86 °F	50 – 60 %	Transition Ventilation
Week 3 (Days 15–21)	25 – 27 °C	77 – 81 °F	50 – 60 %	Transition / Tunnel
Week 4+ (Day 22–Market)	20 – 22 °C	68 – 72 °F	50 – 65 %	Tunnel Cooling (As needed)

Table 1: Recommended Brooder & Housing Environmental Targets

3. Nutrition & Feed Management

Feed represents 60–70% of total poultry production costs. Providing balanced nutritional formulations tailored to specific growth stages is vital for performance.

Feed Specifications & Schedule

- Broiler Starter (Days 0–10):** High protein (22–23% CP), crumble format to promote early gut development and rapid organ growth.

- **Broiler Grower (Days 11–24):** Balanced energy/protein (20–21% CP), mini-pellet format to support skeletal frame and muscle development.
- **Broiler Finisher (Days 25–Market):** High energy, 18–19% CP pellet format designed to maximize muscle yield and weight gain.
- **Layer Phase Feeds:** Formulated with elevated calcium levels (3.5–4.2%) and phosphorus balance for strong eggshell formulation.

Water Quality & Management

Water is the most critical nutrient. Ensure fresh, clean, microbial-free water is available 24/7. Flow rates should be tested weekly (20–30 ml/min for chicks, 80–100 ml/min for adult broilers). Flush water lines regularly to eliminate biofilm buildup.

4. Flock Health & Vaccination Schedule

Preventative vaccination schedule must be strictly executed according to local endemic disease risk profiles. Consult a licensed poultry veterinarian for specific regional adaptations.

Age	Target Disease	Vaccine Type	Administration Method
Day 1 (Hatchery)	Marek's Disease & IB / ND	Live / HVT Vector	Subcutaneous / In-Ovo / Spray
Day 7 – 10	Newcastle Disease (ND) + IB	Live (B1 / LaSota)	Drinking Water or Eye Drop
Day 14	Infectious Bursal Disease (Gumboro)	Intermediate Plus Live	Drinking Water
Day 21	Gumboro Booster (If required)	Intermediate Live	Drinking Water
Day 28	Newcastle Disease Booster	Live (LaSota) / Killed	Drinking Water or Injection

Table 2: Standard Broiler Vaccination Protocol

5. Daily Management Checklist & Monitoring

Routine inspection of the flock should occur at least 3–4 times daily. Farm managers should use the following structured checklist during routine barn walks:

- **Morning Inspection (06:00):**
 - Check temperature, humidity, and ventilation controls.
 - Inspect drinker lines for leaks or blockages; check feed line operation.
 - Record overnight mortality and cull weak/sick birds.
- **Mid-Day Inspection (12:00):**
 - Observe bird distribution and behavioral activity (panting, huddling).
 - Check feeder pan fill levels and adjust hopper heights.
- **Evening Inspection (18:00):**
 - Ensure automatic light timers are set correctly.
 - Verify water tank capacity and supplement doses if medicating.
 - Perform final perimeter security check.

6. Record Keeping & Performance Indicators

Accurate record keeping enables tracking of key performance indicators (KPIs) and identifies management issues early. Key metrics include:

- **Feed Conversion Ratio (FCR):** $\text{Total Feed Consumed (kg)} / \text{Total Live Weight Gained (kg)}$. Target: < 1.55 for broilers.
- **European Production Efficiency Factor (EPEF):** $(\text{Livability \%} \times \text{Body Weight [kg]}) / (\text{Age [days]} \times \text{FCR}) \times 100$. Target: > 380.
- **Mortality Rate:** $\text{Total Dead Birds} / \text{Initial Flock Size} \times 100$. Target: < 3–4% total cycle mortality.